

Bayes' Formula

When we make decisions involving investments, we often start with viewpoints based on our experience and knowledge. These viewpoints may be changed or confirmed by new knowledge and observations. **Bayes' formula** is a rational method for adjusting our viewpoints as we confront new information. Bayes' formula and related concepts are used in many business and investment decision-making contexts.

Bayes' formula makes use of the total probability rule:

$$P(A) = \sum_n P(A \cap B_n). \quad (7)$$

To review, that rule expresses the probability of an event as a weighted average of the probabilities of the event, given a set of scenarios. Bayes' formula works in reverse; more precisely, it reverses the "given that" information. Bayes' formula uses the occurrence of the event to infer the probability of the scenario generating it. For that reason, Bayes' formula is sometimes called an inverse probability. In many applications, including those illustrating its use in this section, an individual is updating his/her beliefs concerning the causes that may have produced a new observation.

Bayes' Formula. Given a set of prior probabilities for an event of interest, if you receive new information, the rule for updating your probability of the event is as follows:

Updated probability of event given the new information

$$= \frac{\text{Probability of the new information given event}}{\text{Unconditional probability of the new information}} \times \text{Prior probability of event.}$$

In probability notation, this formula can be written concisely as follows:

$$P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})} P(\text{Event}). \quad (8)$$

Consider the following example using frequencies—which may be more straightforward initially than probabilities—for illustrating and understanding Bayes' formula. Assume a hypothetical large-cap stock index has 500 member firms, of which 100 are technology firms, and 60 of these had returns of >10 percent, and 40 had returns of ≤10 percent. Of the 400 non-technology firms in the index, 100 had returns of >10 percent, and 300 had returns of ≤10 percent. The tree map in Exhibit 4 is useful for visualizing this example, which is summarized in the table in Exhibit 5.